Mandatory Ambiguity

A Hybrid Human–AI Analysis of Impact Reporting by Benefit Corporations

Joel Gehman, with David S. Lucas and Clara Scheve

GWSB AI in Action Award



THE GEORGE WASHINGTON UNIVERSITY
WASHINGTON, DC

Two fast-growing ways to build a mission into a business

Certified B Corps

10,700+

certified firms in 104 countries

Private certification by B Lab

Together employ 1M+ people

- Ben & Jerry's
- Natura &Co
- Tony's Chocolonely

Both

certified and legally committed

- Patagonia
- Warby Parker
- Danone North America
- Lemonade
- Coursera

Benefit corporations

10,000+

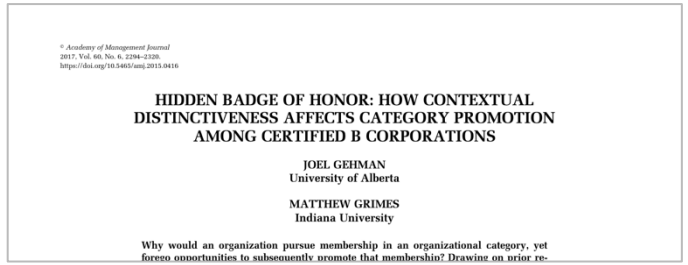
U.S. firms using the legal form

State law in 41 states and D.C.

- Kickstarter
- Anthropic
- OpenAI
- Veeva Systems

B Lab (May 2026); Dorff (2024), Becoming a Public Benefit Corporation; Lucas, Scheve & Gehman (2026). Counts approximate.

Mandatory Ambiguity builds on a decade of B Corp research



HIDDEN BADGE OF HONOR: HOW CONTEXTUAL DISTINCTIVENESS AFFECTS CATEGORY PROMOTION AMONG CERTIFIED B CORPORATIONS

JOEL GEHMAN
University of Alberta

MATTHEW GRIMES
Indiana University

Hidden badge of honor
AMJ 2017



Contents lists available at ScienceDirect

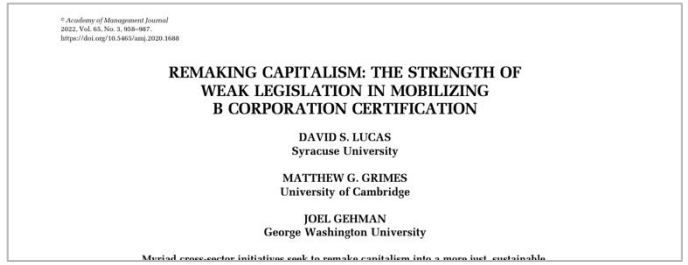
Journal of Business Venturing

journal homepage: www.elsevier.com/locate/jbusvent

Positively deviant: Identity work through B Corporation certification

Matthew G. Grimes, Joel Gehman, Ke Cao

Positively deviant
JBV 2018



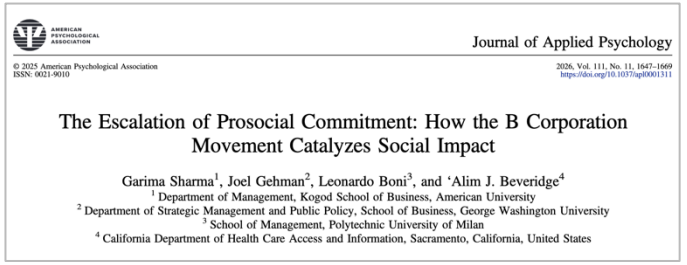
REMAKING CAPITALISM: THE STRENGTH OF WEAK LEGISLATION IN MOBILIZING B CORPORATION CERTIFICATION

DAVID S. LUCAS
Syracuse University

MATTHEW G. GRIMES
University of Cambridge

JOEL GEHMAN
George Washington University

Remaking capitalism
AMJ 2022

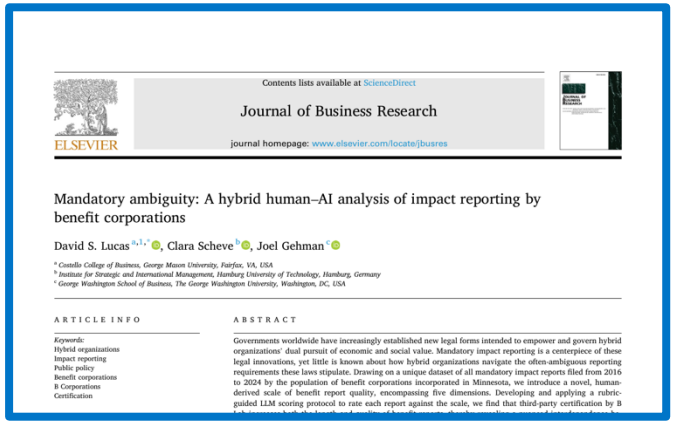


Journal of Applied Psychology

The Escalation of Prosocial Commitment: How the B Corporation Movement Catalyzes Social Impact

Garima Sharma, Joel Gehman, Leonardo Boni, and 'Alim J. Beveridge

Prosocial commitment
JAP 2026



Contents lists available at ScienceDirect

Journal of Business Research

journal homepage: www.elsevier.com/locate/jbusres

Mandatory ambiguity: A hybrid human–AI analysis of impact reporting by benefit corporations

David S. Lucas, Clara Scheve, Joel Gehman

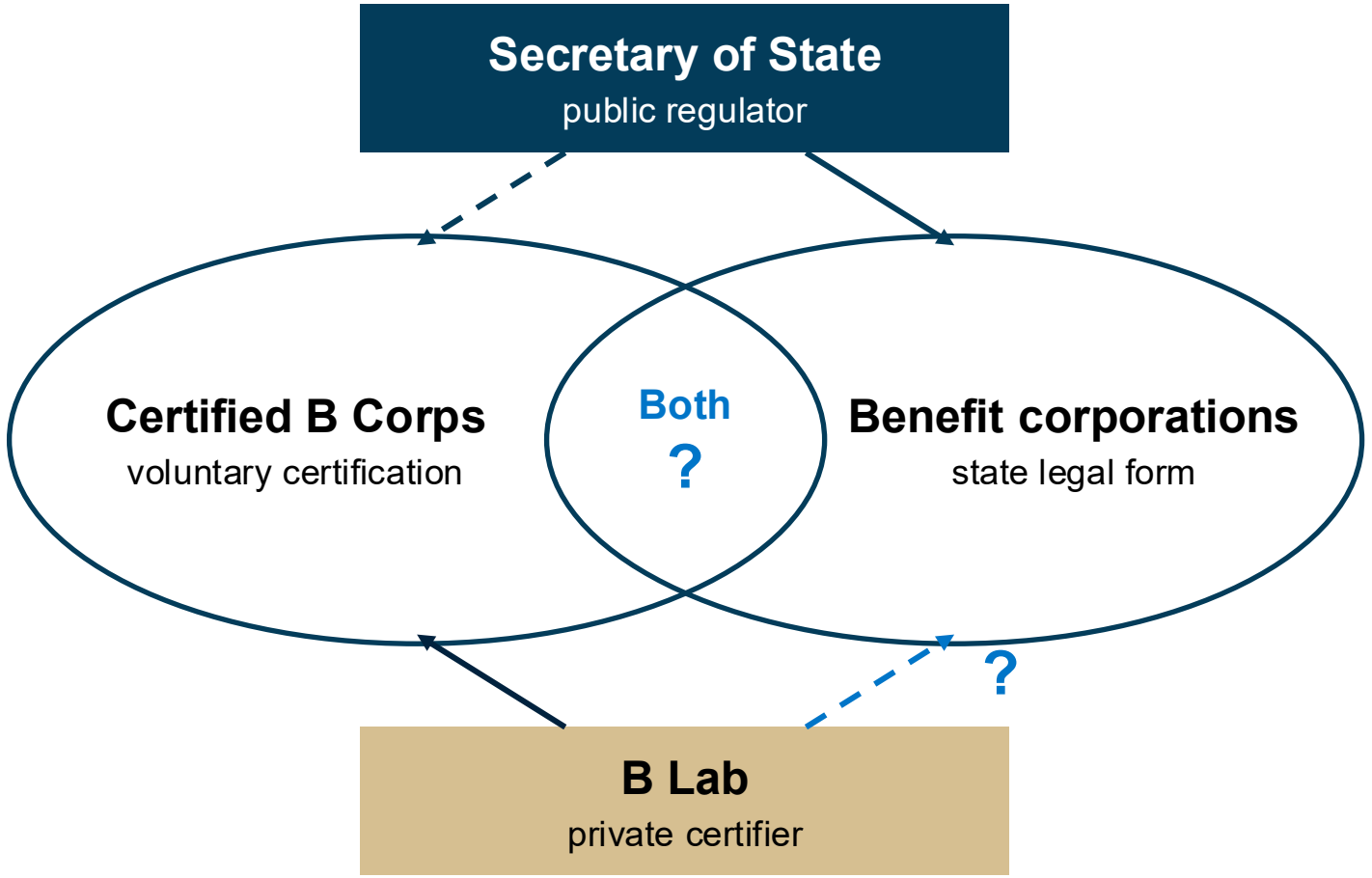
Today: Mandatory ambiguity
JBR 2026

Also: Standing out and fitting in (2018) · Why we care (AMD 2019) · Oxford Bibliographies (2021, 2024) · Generalized specialist (2022)

The puzzle

Benefit corporations, B Corps, and an ambiguous reporting mandate

Two overlapping ways to govern a mission-driven business



Certified B Corp. Voluntary; requires 80+ points on B Lab's B Impact Assessment and a fee.

Benefit corporation. A legal form in 41 states and D.C.; must file an annual benefit report.

Our question. Does B Lab shape the reports the state requires?

Solid arrows: direct governance; dashed: weaker or possible influence. Adapted from Figure 1 in Lucas, Scheve & Gehman (2026); see also Lucas, Grimes & Gehman (2022a).

The law requires a report but says little about what goes in it

Minnesota's definition of a "third-party standard"

"(1) a publicly available standard or guideline for defining, reporting, and assessing the performance of a business enterprise as a social or benefit corporation; (2) promulgated by a natural person or an organization; and (3) that is independent of the public benefit corporation."

- **No floor.** No required length, format, or content.
- **No B Lab.** The statute never mentions it.
- **Only GBCs** (general benefit corporations) must name a standard at all.
- **Weak enforcement** is the norm; filing rates can be near zero.

Minn. Stat. § 304A.021. Murray (2015a, 2021); Aragòn-Correa, Marcus & Vogel (2020).

Three ways B Lab might shape the reports

	Channel	Prediction
H1	Certification	B Corp-certified benefit corporations file longer and better reports
H2	Legal requirement	GBCs, which must name a third-party standard, file longer and better reports
H3	Spillover	Uncertified firms that cite B Lab or its free tools file longer and better reports

Each hypothesis has two parts: (a) report length and (b) report quality.

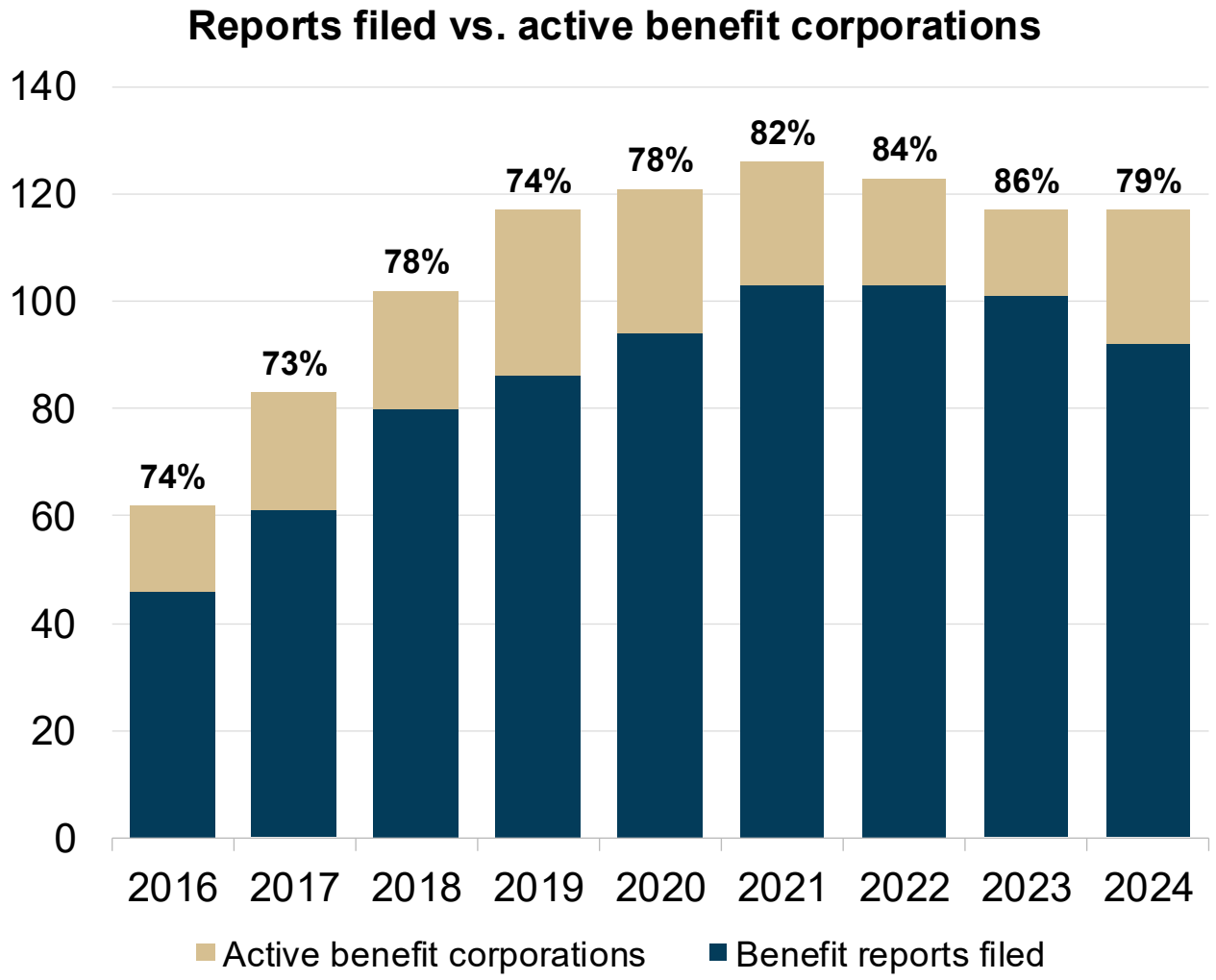
Minnesota posts every report

288 benefit corporations, 2015–2023

766 reports, all scanned PDFs

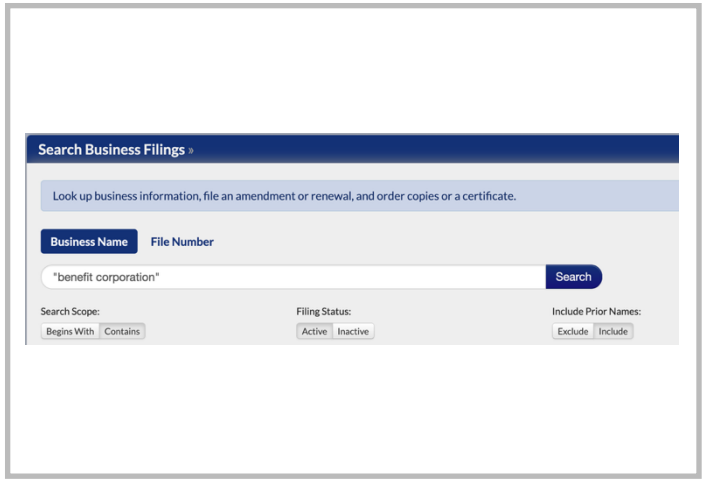
200 firms filed at least once

73–86% filed each year, versus near zero in some states

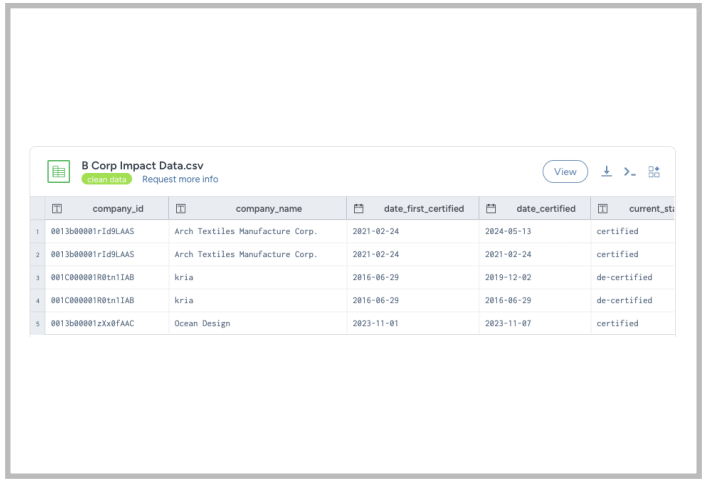


Adapted from Figure 2. Compare Murray (2015a); Verheyden (2018).

We rebuilt the population from public records







1 Search the state registry

Names containing GBC or SBC: 288 benefit corporations

2 Download every posted report

766 annual benefit reports, 2016–2024

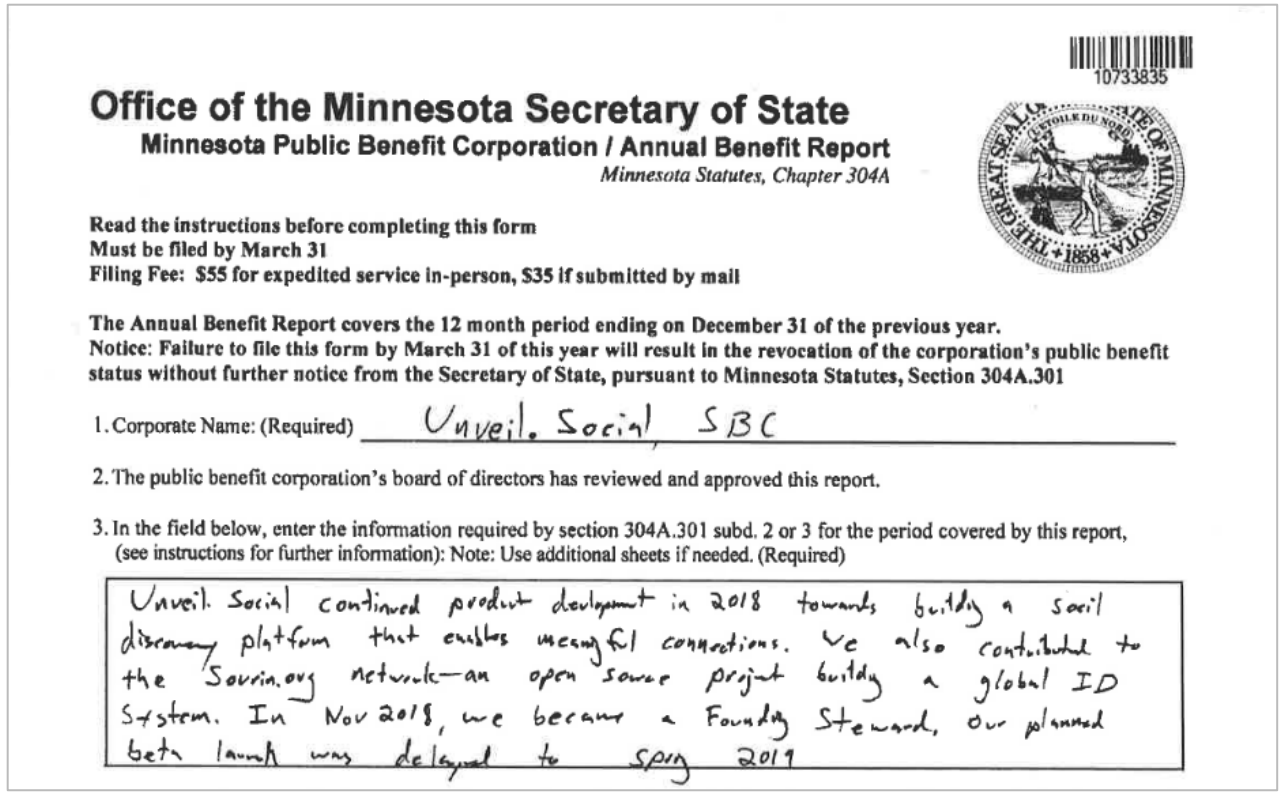
3 Match to B Lab's certified list

20 Certified B Corps, 19 of which filed reports

Population, not sample: to our knowledge, the first such dataset of benefit corporations and their reports

Minnesota Secretary of State business filings and annual benefit reports; B Lab's Certified B Corporations dataset (data.world). Appendix A, Table A1.

From handwritten notes to 80 pages



10733835

Office of the Minnesota Secretary of State
Minnesota Public Benefit Corporation / Annual Benefit Report
Minnesota Statutes, Chapter 304A

Read the instructions before completing this form
Must be filed by March 31
Filing Fee: $55 for expedited service in-person, $35 if submitted by mail

The Annual Benefit Report covers the 12 month period ending on December 31 of the previous year.
Notice: Failure to file this form by March 31 of this year will result in the revocation of the corporation's public benefit status without further notice from the Secretary of State, pursuant to Minnesota Statutes, Section 304A.301

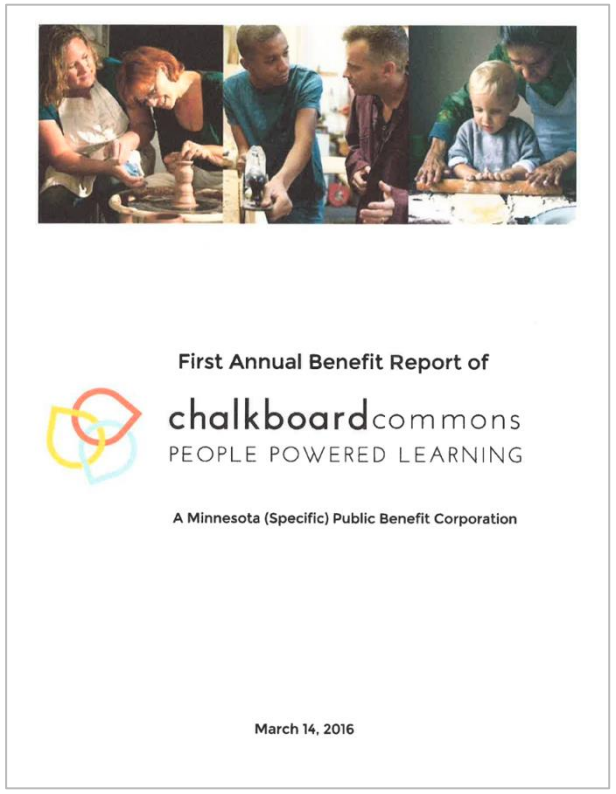
1. Corporate Name: (Required) Unveil. Social SBC

2. The public benefit corporation's board of directors has reviewed and approved this report.

3. In the field below, enter the information required by section 304A.301 subd. 2 or 3 for the period covered by this report, (see instructions for further information): Note: Use additional sheets if needed. (Required)

Unveil. Social continued product development in 2018 towards building a social discovery platform that enables meaningful connections. We also contributed to the Sovrin.org network—an open source project building a global ID System. In Nov 2018, we became a Founding Steward. Our planned beta launch was delayed to Spring 2019

Some filings: a few lines written into the state's form



First Annual Benefit Report of

chalkboardcommons
PEOPLE POWERED LEARNING

A Minnesota (Specific) Public Benefit Corporation

March 14, 2016

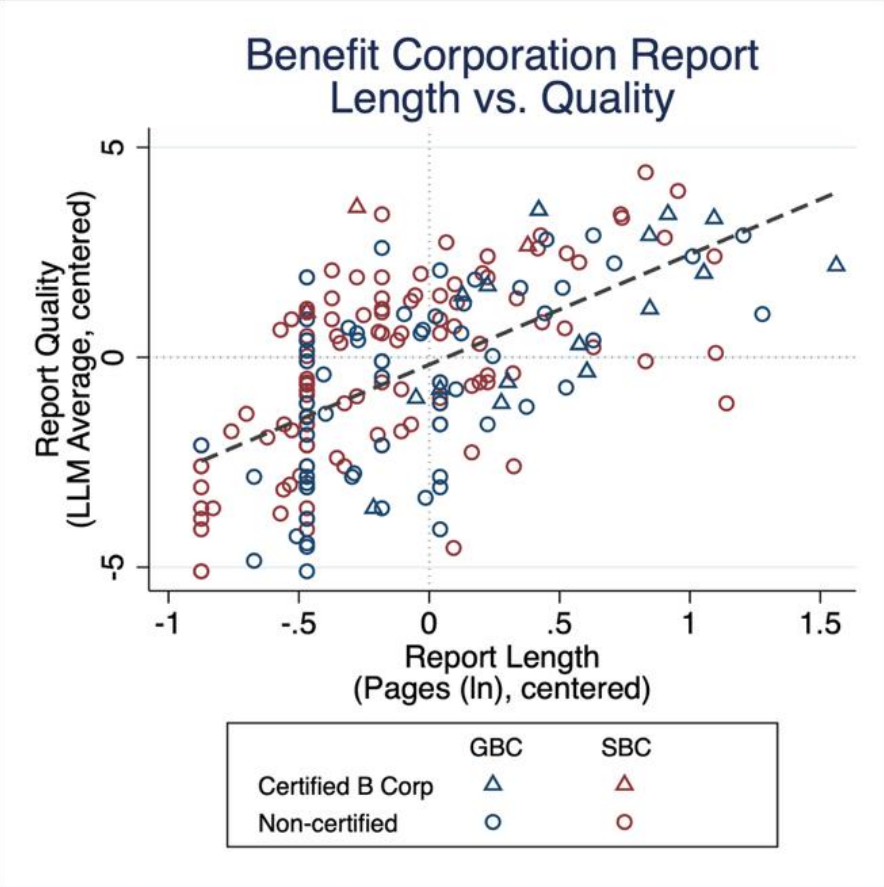
Others: professional reports with metrics and photos

Public benefit reports filed with the Minnesota Secretary of State. All were scanned images; we ran OCR and hand-transcribed what OCR could not read.

Our hybrid human–AI method

Humans define quality, LLMs apply it at scale, humans check the work

The measurement problem: 766 reports and no yardstick for quality



0 existing scales for benefit report quality

0.52 length–quality correlation; a third of firms fall off the diagonal

3,830 scores needed: 766 reports × 5 dimensions, each with a reason

Why AI: apply one human-built rubric the same way to every report, with reasons we can audit

Figure C2, appendix C: firm averages, mean-centered.
Longer disclosures can obscure: Loughran & McDonald (2014); Melloni et al. (2017).

People decide what counts; LLMs do the repetitive scoring



Code reports inductively; write the rubric with anchors

Two models score every report, each score with a reason

Review all scores; re-run cases where models disagree

Deductive, not inductive. Many studies let an LLM find themes on its own. We did the reverse, so interpretive authority stayed with us and the models worked as structured assistants.

Inductive uses: Hayes (2025); Naeem, Smith & Thomas (2025). LLMs as structured assistants: Glaser & Gehman in Kulkarni et al. (2024). Human–AI entanglement: Gehman, Glaser & Merritt (2025).

Four phases, each ending in a validated output

Phase	Who	What happened	Output
1. Build the framework	Human	Hand-coded 119 reports; 114 codes grouped into 5 dimensions	10-page rubric
2. Calibrate	Human + LLMs	Tested 7 model variants; refined the prompt	Validated prompt and 2 models
3. Score everything	GPT-4o and Claude Sonnet 4	All 766 reports, in batches of 5 to 10	Model agreement r = 0.73
4. Quality control	Human	Reviewed all scores; re-ran 79 with 4+ points of disagreement	Model agreement r = 0.83

Model agreement is the correlation between the two models' total scores. Adapted from Table 1.

Phase 1: a five-dimension rubric, built by hand before any AI

Dimension	0: Low	1: Medium	2: High
Mission clarity	Vague or generic	Stated but broad	Concrete goals, clear beneficiaries
Stakeholder engagement	Little or none	Ad hoc or one-way	Real collaboration and partnerships
Impact evidence	Activities only	Some metrics, little context	Specific, verified metrics
Transparency and candor	Reads like marketing	Some issues noted	Frank about failures and lessons
Future commitments	No plans	Vague intentions	Measurable targets and timelines

Total score 0–10. Coded from 20 reports, refined on 99 more, and informed by prior research.

Table 2 in the article; Corbin & Strauss (1990).

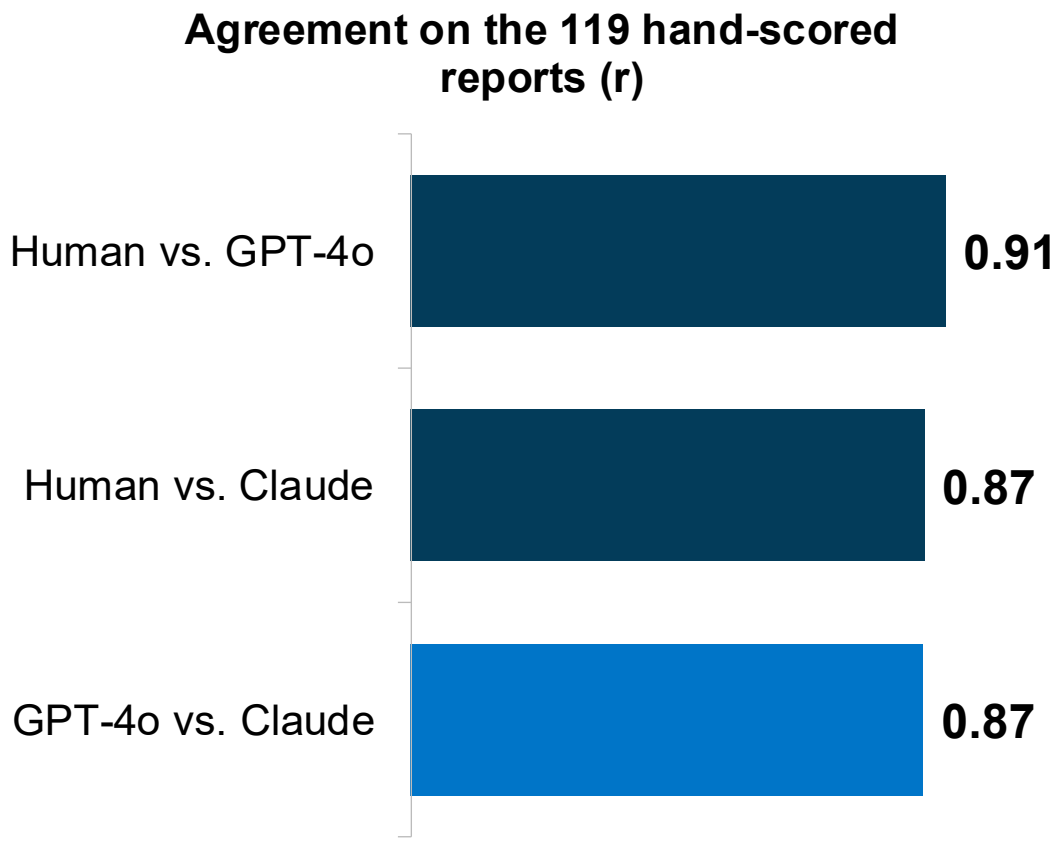
Phase 1: anchor quotes make the rubric concrete

0 Activities only	1 Thin metrics	2 Specific evidence
"First Aid for Nurses delved into social media this past year on Pinterest…"	*"Since 2010, we have improved across all B Corp metrics by double digit points…"*	*"Contributed $13,259.79 and volunteered more than 805 hours…"*
First Aid for Nurses SBC, 2017	CR Building Performance Specialists GBC, 2021	Dark Clouds GBC, 2023

Quotations from public benefit reports filed with the Minnesota Secretary of State; see Table 2.

Phase 2: test several models against human scores

- **Tested:** ChatGPT-4o, GPT-4.1, GPT-4.5, o3, Claude Opus 3, Opus 4, and Sonnet 4.
- **Chosen:** ChatGPT-4o and Claude Sonnet 4 tracked human scores most closely.
- **Not always the newest:** larger or newer models were not automatically better scorers.
- **Zero-shot:** no pre-scored example reports, only the prompt and the full rubric.



Correlations from Table 1. On zero-shot, rubric-embedded classification: Chae & Davidson (2025).

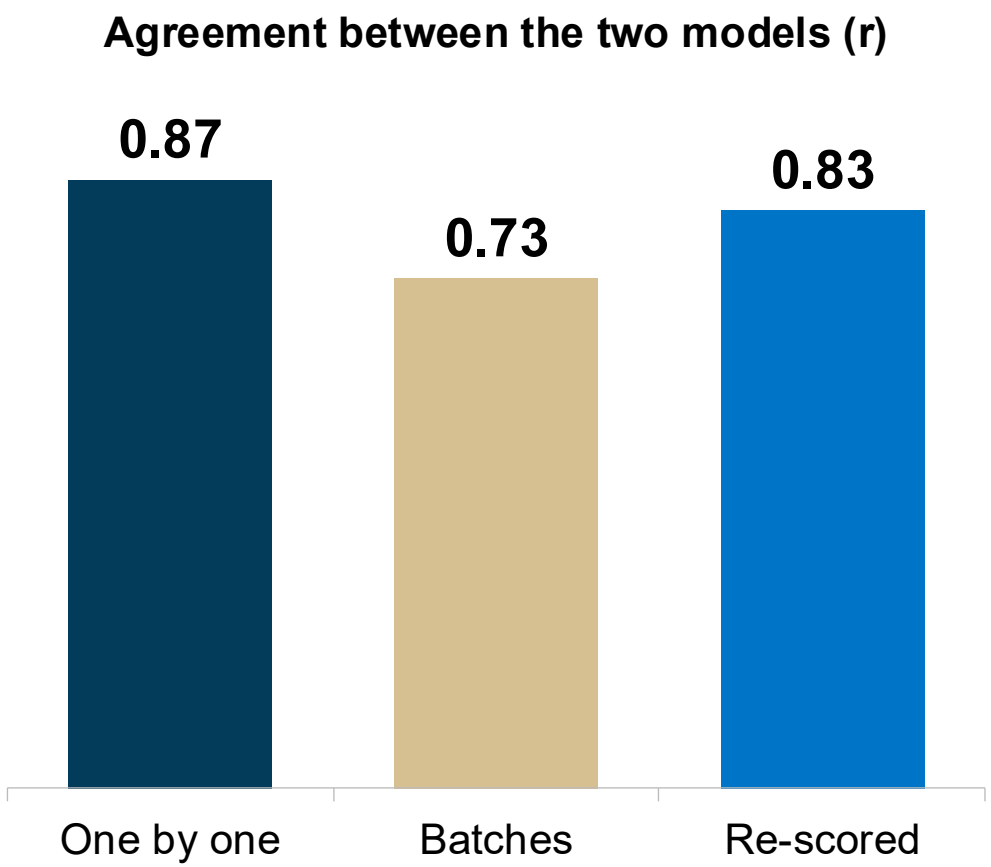
Phase 2: a prompt with strict rules and known mistakes

Part	Excerpt from the prompt
Role	"You are an expert evaluator of public benefit corporation annual reports."
Six rules	"Be Highly Critical… Score only what is actually demonstrated with evidence."
Errors to avoid	"Don't confuse business development with public benefit."
Output	A table: a short reason and a 0–2 score per dimension, then the total

The full 10-page rubric followed the prompt in every session.

Full prompt text: Table B1.

Phases 3–4: scaling cost agreement; re-scoring won it back



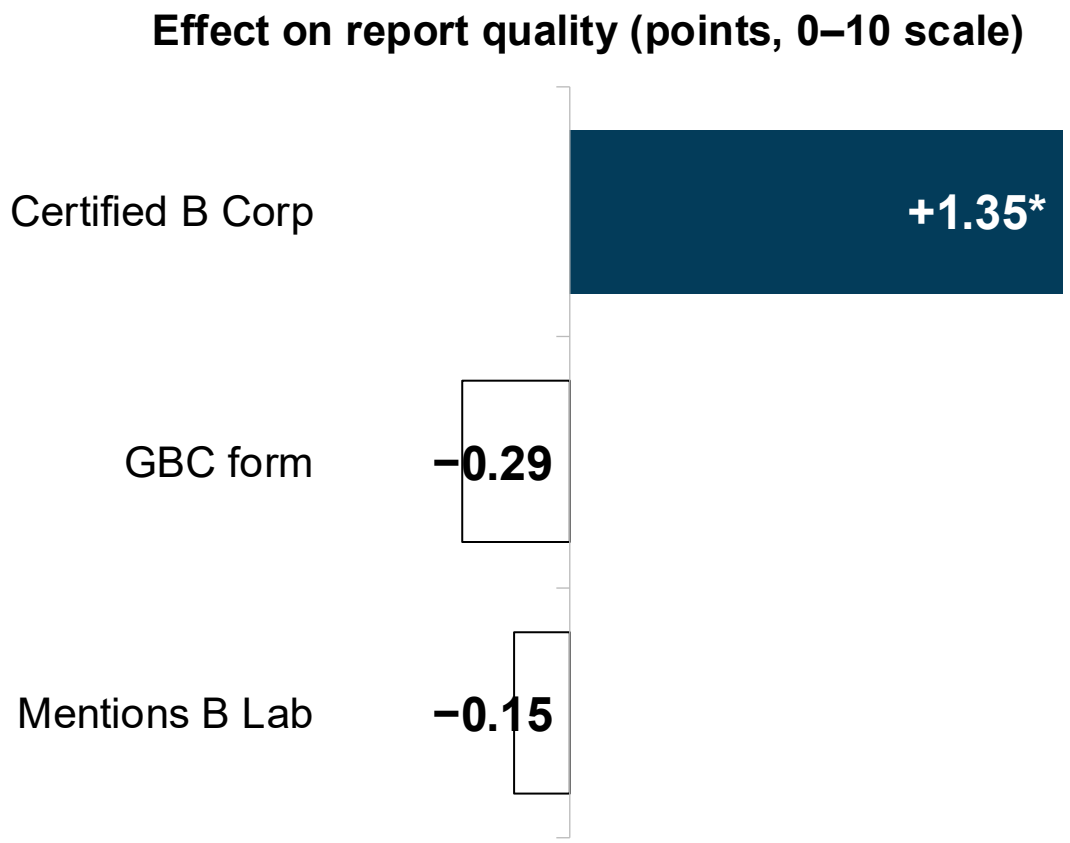
- **Batch.** Prompt and rubric repeated at the top of every batch.
- **Clean.** A person checked all 1,532 scores.
- **Flag.** 79 reports (10%) where models differed by 4+ points.
- **Re-score.** Each re-run alone; largest gap fell to 3 points.
- **Likely cause:** models attend less to the middle of long inputs.

Table 1 and Appendix B. “Lost in the middle”: An et al. (2024). On validating and overseeing AI annotation: Pangakis, Wolken & Fasching (2023); Morgan (2023).

What we found

Certification shapes the substance of mandatory reports

Certification predicts better reports; the legal form and B Lab mentions do not



H1 supported. B Corps' reports are about 40% longer and score 1.35 points higher (about half an SD).

H2 half supported. GBCs write longer reports, but not better ones.

H3 not supported. Mentioning B Lab without certifying changes neither.

Table 4; * p < 0.05; GBC and B Lab effects not significant. OLS with year and sector effects, firm-clustered errors. Results holds with firm fixed effects (Bruno, Magazzini & Stampini, 2020) and matching.

Certified firms stand out on evidence, candor, and future commitments

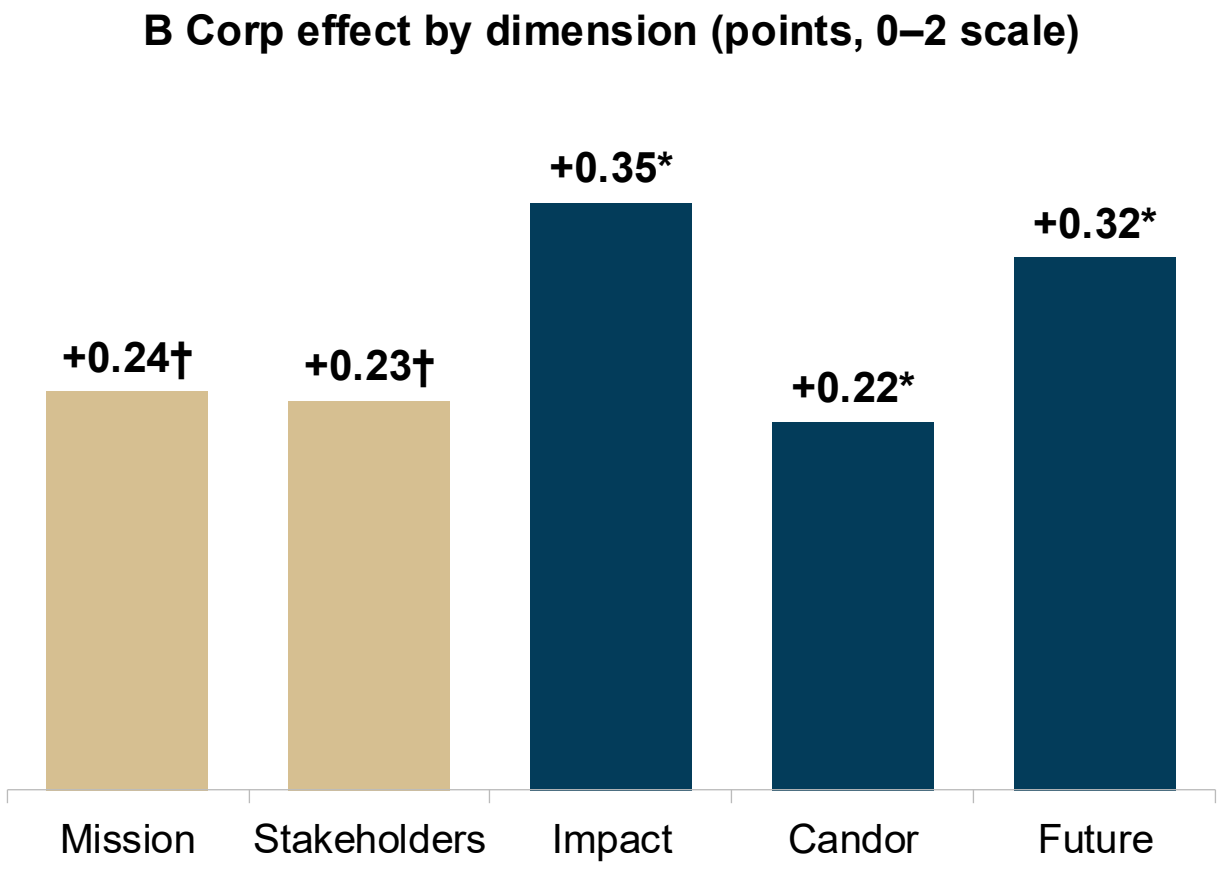
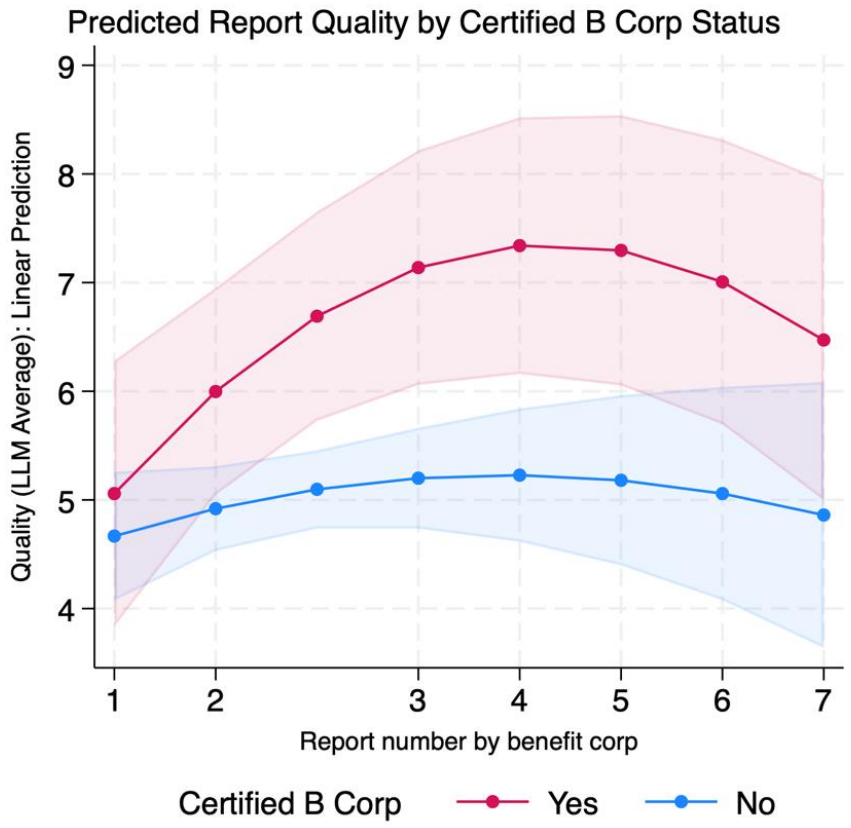


Table 6. * p < 0.05; † p < 0.10.

Saying vs. showing. Every benefit corporation states a mission. Certified firms are more likely to show how they pursue it: measured impact, candor about shortfalls, and specific targets.

Certified firms improve faster with each report they file



- **Steeper curve.** Predicted quality rises faster across successive reports for Certified B Corps.
- **A big gap by report four:** about 2.35 points on the 10-point scale, nearly a full standard deviation.
- **More upward moves.** 37% of B Corps moved up a quality tier from first to last report, vs. 29% of others.

Figure C5 and Tables C1–C2. Predicted values with 95% confidence intervals; exploratory analysis.

Some implications

Policy	Practice	Professors
Enforcement gets reports filed, not good reports. Some ambiguity may be the price of passing these laws.	Certification gives firms a real framework for better reports. Borrowing B Lab's tools without certifying does not.	A reusable quality scale for impact reports, and a documented hybrid human–AI method.

The big idea: third-party organizations give meaning to ambiguous prosocial laws.



You might try it too

A recipe, the principles behind it, and where it fits

A recipe you can borrow

1 Code a sample by hand

Stop when your categories stop changing

2 Write the rubric

Define each score level, with anchor quotes

3 Embed it in a strict prompt

List the mistakes you see the model make

4 Test models on your scores

Newest isn't always best

5 Run two, compare, re-score

Keep inputs short; average the two

6 Review, record, disclose

Log models, dates, and prompts

Where it could fit at GWSB

Research	Teaching	Service
10-Ks, ESG reports, earnings calls, open-ended survey items, interview transcripts	A first pass against your grading rubric, with you making the call	Sorting hundreds of comments from surveys, evaluations, and program reviews

Caution: we used public documents. Student work and confidential data need FERPA, IRB, and GW AI-policy review first.

Even as models advance, these principles are still state of the art

1

Corroborate across models

Disagreement between models is your early-warning system

2

Corroborate against people

A hand-scored benchmark is the yardstick, before and after scaling

3

Keep judgment human

People write the rubric and own the final scores

See also Pangakis, Wolken & Fasching (2023); Chae & Davidson (2025).

Part of a broader line of my work on AI


JOURNAL OF MANAGEMENT STUDIES

Check for updates

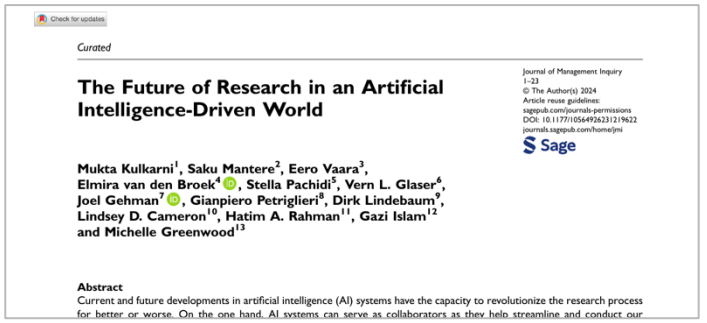
Journal of Management Studies 61:6 September 2024
doi:10.1111/joms.13033

POINT
COUNTERPOINT

Organizations as Algorithms: A New Metaphor for Advancing Management Theory

Vern L. Glaser[a], Jennifer Sloan[a] and Joel Gehman[b]

Organizations as algorithms
Journal of Management Studies, 2024


Check for updates

Curated

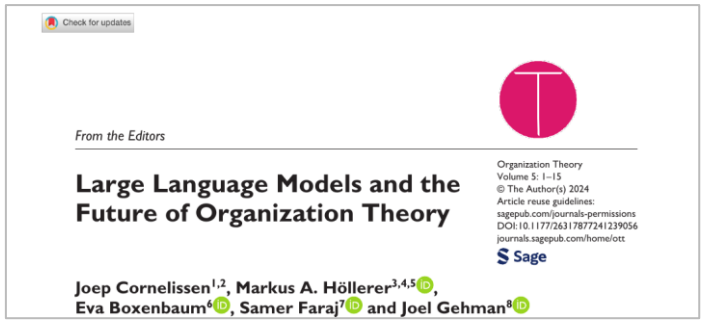
The Future of Research in an Artificial Intelligence-Driven World

Journal of Management Inquiry
1–23
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Sage

Mukta Kulkarni[1], Saku Mantere[2], Eero Vaara[3], Elmira van den Broek[4], Stella Pachidi[5], Vern L. Glaser[6], Joel Gehman[7], Gianpiero Petriglieri[8], Dirk Lindebaum[9], Lindsey D. Cameron[10], Hatim A. Rahman[11], Gazi Islam[12] and Michelle Greenwood[13]

Abstract
Current and future developments in artificial intelligence (AI) systems have the capacity to revolutionize the research process for better or worse. On the one hand, AI systems can serve as collaborators as they help streamline and conduct our

Research with AI
Journal of Management Inquiry, 2024


Check for updates

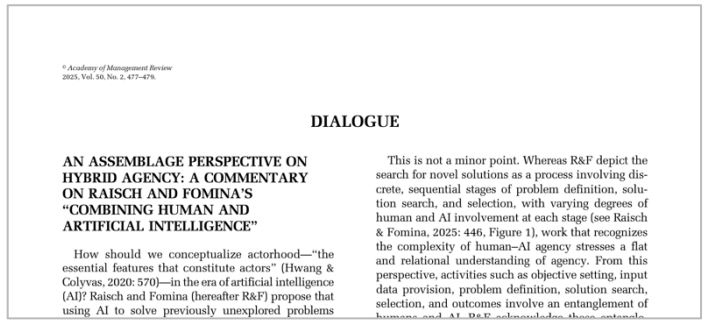
From the Editors

Large Language Models and the Future of Organization Theory

Organization Theory
Volume 5: 1–15
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DOI: 10.1177/26317877241239056
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Joep Cornelissen[1,2], Markus A. Höllerer[3,4,5], Eva Boxenbaum[6], Samer Faraj[7] and Joel Gehman[8]

LLMs and organization theory
Organization Theory, 2024


Academy of Management Review
2025, Vol. 50, No. 2, 477–479.

DIALOGUE

AN ASSEMBLAGE PERSPECTIVE ON HYBRID AGENCY: A COMMENTARY ON RAISCH AND FOMINA'S "COMBINING HUMAN AND ARTIFICIAL INTELLIGENCE"

How should we conceptualize actorhood—"the essential features that constitute actors" (Hwang & Colyvas, 2020: 570)—in the era of artificial intelligence (AI)? Raisch and Fomina (hereafter R&F) propose that using AI to solve previously unexplored problems

This is not a minor point. Whereas R&F depict the search for novel solutions as a process involving discrete, sequential stages of problem definition, solution search, and selection, with varying degrees of human and AI involvement at each stage (see Raisch & Fomina, 2025: 446, Figure 1), work that recognizes the complexity of human–AI agency stresses a flat and relational understanding of agency. From this perspective, activities such as objective setting, input data provision, problem definition, solution search, selection, and outcomes involve an entanglement of

Hybrid agency as assemblage
Academy of Mgmt. Review, 2025


AI and Ethics (2026) 6:576
https://doi.org/10.1007/s43681-026-01439-y

RESEARCH

Beyond one-size-fits-all: how adaptability and autonomy shape responsible AI governance

Mohammad Hossein Jarrahi[1] · Vern L. Glaser[2] · Joel Gehman[3]

Received: 25 July 2026 / Accepted: 27 September 2026
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Abstract
Artificial intelligence (AI) governance is a critical safeguard against risk and ethical harm. However, organizations need clearer guidance on how oversight should vary across an AI portfolio. Building on characteristics recognized in scholarly

Responsible AI governance
AI & Ethics, 2026


The Token-Based View of Work
Merritt, Gehman & Rissing

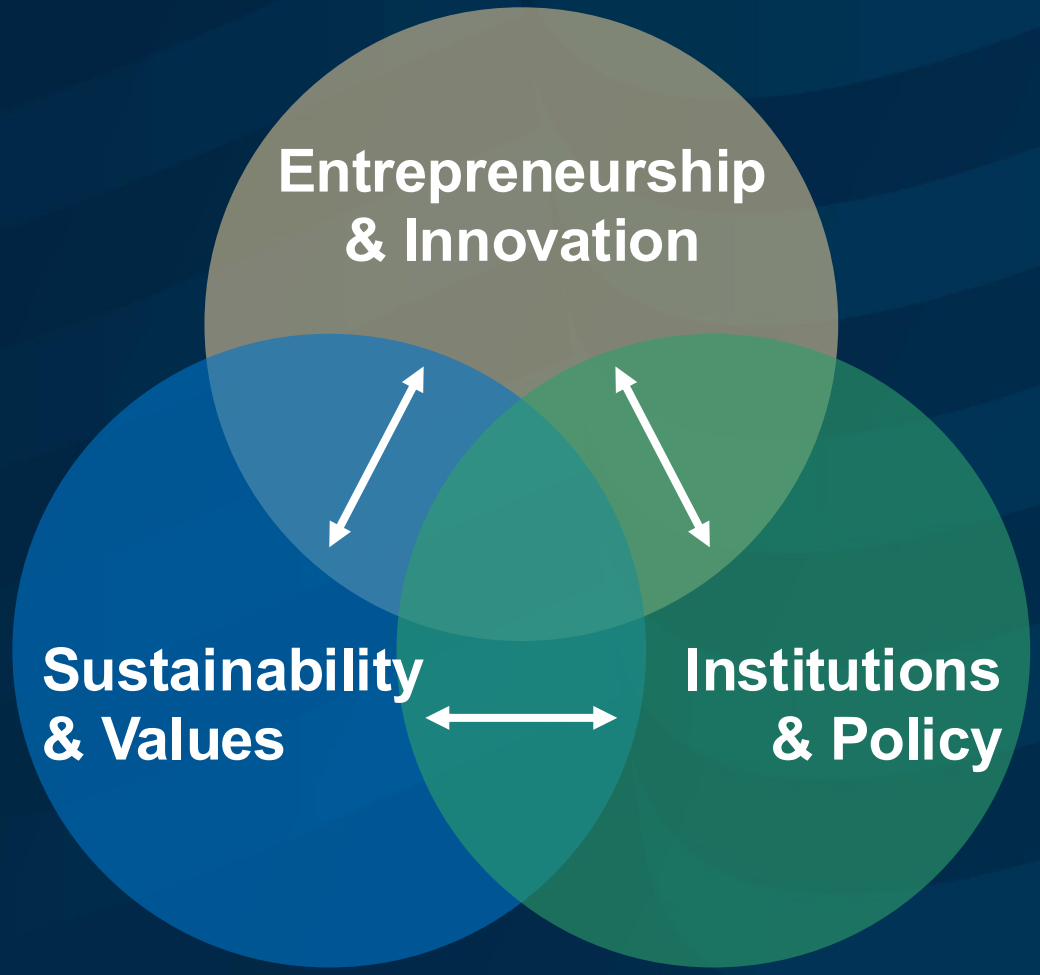
The token-based view of work
Organization Science, under review

How can business be a force for good?

Joel Gehman

Lindner-Gambal Professor of Business Ethics

My research examines how businesses and other organizations can help tackle grand challenges related to sustainability, through strategic practices, technological innovation, and institutional change.



75+

Peer-reviewed articles & book chapters

$5M+

In research funding awarded

12,000+

Citations to my publications

Citations: Google Scholar, Nov. 2026

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Lucas, D. S., Grimes, M. G., & Gehman, J. (2022a). Remaking capitalism: The strength of weak legislation in mobilizing B Corporation certification. Academy of Management Journal, 65(3), 958–987.

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